XINGCHEN SHA

(+1) 3475393252 • xingchen.sha@u.northwestern.edu • Web: http://xingchen-sha.com

EDUCATION

Ph.D. in Computer Science — May 2030 (expected)
Northwestern University, Evanston, United States — Advisor: Edith Elkind

B.A. in Computer Science — May 2025
Columbia University, New York, United States — GPA: 4.03

B.S. in Computer Science — May 2025
City University of Hong Kong, Kowloon Tong, Hong Kong — GPA: 4.01

EXPERIENCE

Columbia University: Research Assistant — October 2023 – May 2025

- Collaborator in the NSF CSIRO-funded project Fair Sequential Collective Decision Making
- Researched automatic algorithms design with LLMs
- Researched group-fairness in multi-period facility location problems
- Advisor: Haris Aziz, Hau Chan, Toby Walsh, Lirong Xia

City University of Hong Kong: Research Assistant — June 2022 – April 2023

- Researched facility location games with capacity constraints
- Researched multi-stage facilty location games in dynamic settings
- Researched truthful mechanism design with optional preferences
- Advisor: Hau Chan, Vincent Chau, Ken Fong, Minming Li, Pinyan Lu, Jialin Zhang

ACTIVITY

Columbia University: Student Helper — September 2024 – May 2025
Decentralized Finance and Game Theory Seminar

City University of Hong Kong: Spotlight Presenter — June 2023
STEM Carnival cum Student Project Exhibition 2023

City University of Hong Kong: Organizer — May 2023 – September 2023
Hong Kong Secondary School Coding Challenge 2023 (HKSC2023)

City University of Hong Kong: Student Helper — September 2022 – September 2023
CityU Theory Seminar

City University of Hong Kong: Organizer — May 2022 – September 2022
Hong Kong Secondary School Coding Challenge 2022 (HKSC2022)

SERVICES

City University of Hong Kong: College of Engineering Student Chapter Panel — November 2022 - May 2025
City University of Hong Kong: Freshman Mentor — September 2022 - December 2022

SELECTED HONORS & AWARDS

College of Computing College Medal — January 2026
GS Honor Society — February 2025
Upsilon Pi Epsilon — September 2024
Dean's List at Columbia University — September 2023 – January 2025
VTech Group of Companies Scholarship — June 2023
Dean's List at City University of Hong Kong — September 2020 – May 2023
The College of Engineering Dean's Scholarship — April 2023

HKSAR Government Scholarship February 2023
CityU Scholarship February 2022

PUBLICATIONS (* ALPHABETICAL ORDER)

Large Language Models for Indivisible Participatory Budgeting
Nguyen Thach, **Xingchen Sha**, Hau Chan, AAMAS 2026

Mechanism Design for Facility Location Problems with Capacity Constraints in Bounded Location Space
Xingchen Sha, Hau Chan, Vincent Chau, Ken Fong, Minming Li, Wai Lun Lo, ECAI 2025

Strategyproof Mechanism for Two Heterogeneous Facilities with Constant Approximation Ratio *
Minming Li, Pinyan Lu, **Xingchen Sha**, Yuhao Yao, Jialin Zhang, Games and Economic Behavior

Facility Location Games with Optional Preferences: A Revisit
Xingchen Sha, Shuyu Bao, Hau Chan, Vincent Chau, Ken Fong, Minming Li, AAAI 2025

Randomized Strategyproof Mechanisms for Multi-stage Facility Location Problem with Capacity Constraints
Ken Fong, **Xingchen Sha**, Hau Chan, Vincent Chau, Wai Lun Lo, IJTCS-FAW 2024

Group Fairness in Multi-period Mobile Facility Location Problems *
Haris Aziz, Hau Chan, **Xingchen Sha**, Toby Walsh, Lirong Xia, AAMAS 2025 (Extended Abstract), ADT 2024 (Abstract)

SKILLS

Programming Languages:
C/C++, Python, Java, SQL, Bash, HTML

Professional Applications:
Latex, Matlab, PyTorch, TensorFlow, Git